MILITARY SPECIFICATION

PUMPING UNIT, FRAME MOUNTED, 1-1/2 INCH, ELECTRIC MOTOR DRIVEN, 35 GPM AT 25 FEET TOTAL HEAD

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an electric motor driven pumping unit and associated hose and fitting. The pump has a capacity of 35 GPM at 25 feet total head.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4320
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SPECIFICATIONS

MILITARY

MIL-V-173	- Varnish, Moisture and Fungus Resistant (For Treatment of Communications Electronics and Associated Equipment).
MIL-H-370	- Hose and Hose Assemblies, Non-metallic, Rubber, Liquid Fuel, Non-collapsible.
MIL-P-514	- Plate, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power Driven, for Water, Packaging of.
MIL-H-11588	- Hose Assembly, Rubber, Synthetic, Liquid Petroleum Fuels, Dispensing, Collapsible.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-N-52110	- Nozzles, Fuel & Oil Servicing, Non-automatic Shutoff and Nozzles, Fuel Servicing, Automatic Shutoff.
MIL-C-62122	- Cable Assembly, Slave, Electrical Connectors, Plug and Receptacle.

STANDARDS

FEDERAL

FED-STD-H28	- Screw Thread for Federal Service.

MILITARY

MIL-STD-130	– Identification Marking of U.S. Military Property.
MIL-STD-454	– Standard General Requirements for Electronic Equipment.
MIL-STD-461	– Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-810	– Environmental Test Methods and Engineering Guidelines.
MIL-STD-889	– Dissimilar Metals.
MIL-STD-1472	– Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	– Noise Limits for Army Materiels.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

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DRAWINGS

13220E1127 - Ground Cable Assembly.

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the non-Government documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

AMERICAN WELDING SOCIETY (AWS)

D1.1 - Structural Welding Code - Steel
D1.2 - Structural Welding Code - Aluminum

(Application for copies should be addressed to the American Welding Society, Inc., 550 N.W. LeJeune Road, P.O. Box 351040, Miami, FL 33125.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 26 - Aluminum Alloy Sand Castings, Spec for.
B 210 - Aluminum-Alloy Drawn Seamless Tubes.
B 221 - Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Shapes, and Tubes.
B 584 - Copper Alloy Sand Castings for General Application, Spec for.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 122 East 42nd Street, New York, NY 10017.)

NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

MG-1 - Motors and Generators.

(Application for copies should be addressed to the National Electrical Manufacturers Association, 815 15th Street, Suite 438, Washington, DC 20005.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

No. 70 - National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

UNDERWRITERS Laboratory (UL)

UL 674 - Electric Motors and Generators for use in Laboratories, Inc.

(Application for copies should be addressed to Underwriters Laboratories, Inc., 333 Pfingsten Road, Northbrook Road, IL 60062.)

SOUTH COAST AIR QUALITY MANAGEMENT DISTRICT

Rule 422 - Usage of Solvents.

(Application for copies should be addressed to South Coast Air Quality Management District, 9150 East Flair Drive, El Monte, CA 91731.)

(Non-Government standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specifications sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pumping unit shall be a self priming centrifugal pump as defined by the Hydraulic Institute with a capacity of 35 GPM at 25 feet total head.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the pumping unit from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumping units conforming to this specification.

3.3 Materials. Materials shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

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3.3.1 Material deterioration prevention and control. The pumping unit shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the pumping unit may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces, and parts incorporated in the pumping unit may be newly fabricated from recovered materials to the maximum extent practicable, provided the pumping unit produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces, and parts shall not be incorporated in the pumping unit.

3.4 Environmental.

3.4.1 Operating temperature. The pumping unit shall perform at rated capacity at any ambient temperature from +125 to -25 °F.

3.4.2 Storage temperature. The pumping unit shall not be damaged by storage in any ambient temperature from +155 to -65 °F.

3.5 Safety. All shafts or rotating parts of the pumping unit shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Adequate shielding may be used to attain the same result. External surfaces shall be free of sharp edges and corners except where sharp edges and corners are necessary and are not detrimental to safety. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Requirement 1 of MIL-STD-454 shall apply.

3.6 Human factors engineering. The pumping unit shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472, as applicable. Special design emphasis shall be given, but not limited to 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design for Maintainability), 5.13 (Hazards and Safety) of MIL-STD-1472.

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3.7 Noise limits. The noise produced by the pump shall conform with MIL-STD-1474. The pump shall meet MIL-STD-1474, 5.1.1, category E requirements when tested in accordance with 4.6.6. Category E requirements may be exceeded if and only if MIL-STD-1474, 5.1.1.2.3 procedures, modified for category E [75 dB(A)], have been persued and documented to the satisfaction of the procuring activity and written permission to exceed the category E limit is obtained from the procuring activity.

3.8 Pump. The pump shall be a self-priming centrifugal (after case has been initially filled) with minimum flow of 35 GPM at 25 feet total head.

3.8.1 Tilted position. The pumping unit shall deliver fuel at its rated capacity when positioned at an angle up to 15 degrees from the horizontal.

3.8.2 Priming. The pumping unit shall prime within two minutes on a 10-foot static suction lift after initial filling of the pump casing. Priming of pump shall not require removal of a pipe plug, pipe cap or use of any tools.

3.8.3 Priming and sampling pump. The pumping unit shall be equipped with a hand piston type pump with spout to be used to collect fuel samples for testing and also to initially fill the pump casing before priming.

3.8.4 Construction.

3.8.4.1 Pump case and volute. The pump case and volute shall be a one piece aluminum casting conforming to ASTM B 26, alloy 356.0-T6. The pump casing shall incorporate a drain receptacle. The casing shall have the suction and discharge ports marked by the terms “INLET” and “OUTLET” cast in the pump casing. A direction arrow shall be cast in the pump casing to indicate the direction of impeller rotation.

3.8.4.2 Intermediate bracket. The pump assembly shall be connected to the motor by a one piece intermediate bracket constructed from 356-T6 aluminum. The intermediate bracket shall incorporate the mechanical seal housing and provide a minimum of two mounting holes to rigidly mount the pump and motor assembly to the frame.

3.8.4.3 Impeller. The impeller shall be fabricated of bronze conforming to ASTM B 584, copper alloy UNS No. C93800.

3.8.4.4 Seals. The pump shall be equipped with self-lubricating mechanical seals that permit operation of the pumping assembly in any operating condition specified herein.

3.8.4.5 Screw threads. Screw threads shall conform to FED-STD-H28,

3.9 Electric motor. The motor shall conform to NEMA MG-1, be direct current (dc, 24 volt, totally enclosed, non-ventilated, water-proof and explosion proof. The motor shall be capable of operating the pump at rated conditions for a continuous period of one hour without overheating or any other failure.

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Motor shall be equipped with a thermostat to sense approaching overheat conditions and shut motor off before damage occurs. The motor shall conform to UL 674 for class I, groups C and D and class II, groups F and G.

3.9.1 Explosion-proof. The electric motor assembly complete with all switches shall be explosion-proof in accordance with National Fire Protection Association Code NFPA No. 70, class 1, Division 1, group D.

3.9.2 Motor protection. The electric motor shall be provided with short circuit and overload protection in accordance with the National Fire Protection Association Code NFPA-70.

3.9.3 Slave cable. The pumping unit shall be equipped with a minimum 15-foot slave cable of sufficient size to safely carry the power required to operate the unit. The cable shall be oil resistant and incorporate a slave connection conforming to MIL-C-62122.

3.10 Protective frame. The unit shall be housed in a tubular aluminum protective frame. The aluminum tubing shall conform to ASTM B 210 or B 221, Alloy G081-T6. The frame shall allow access for operation and maintenance. The frame shall be of sufficient strength to allow the unit to be dropped from a height of four feet. No part of the pumping unit shall extend past the outer edge of the protective frame. The frame shall contain provisions for storing the slave cable. The on-off switch shall be mounted on the frame. The frame shall have two handles or grasp areas to allow for two-soldier transportability.

3.11 Lifting and tiedown devices. The pumping unit shall have lifting/ tiedown devices in combinations or separately to secure the pumping unit to a common carrier or for lifting the unit by military personnel. Lifting and tiedown provisions shall be labeled “LIFT HERE” and “TIE DOWN HERE” respectively.

3.12 Grounding assembly. One ground cable conforming to Dwg 13220E1127 shall be furnished, The pumping unit shall incorporate a ground terminal to be compatible with the ground cable assembly as noted on DWG 13220E1127.

3.13 Hose. The pumping unit shall be furnished with two 25 feet long suction hoses conforming to MIL-H-370 type I, size 5 with class I, style A fittings. The pumping unit shall also be provided with two 25 feet long discharge hoses conforming to MIL-H-11588, type III with fitting as specified above.

3.13.1 Suction strainer. A suction strainer with a maximum of 40 mesh openings shall be furnished. The strainer shall be furnished with a male fitting that will mate with the suction hoses specified in 3.13.

3.13.2 Nozzle. The pump unit shall be furnished with a 1-1/2-inch nozzle conforming to MIL-N-52110. The nozzle shall be adaptable to the discharge hose specified in 3.13.

3.14 Identification. Each pumping unit shall be identified in accordance with MIL-STD-130, on plates conforming to MIL-P-514, type III, composition C,

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grade B, class I. The plates shall be attached to the pumping unit by screws, rivets, or bolts in a place where the plate will be visible and legible but not interfere with the operation and maintenance of the pumping unit.

3.15 Lubrication. The pumping unit shall incorporate bearings of the sealed pre-lubricated types.

3.16 Electromagnetic interference. Unless otherwise specified (see 6.2). the pumping unit shall be equipped for the suppression of electromagnetic interference in accordance with MIL-STD-461 for other than engine-driven equipment.

3.17 Treatment and painting. The portions of the pumping unit normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G, as applicable. Unless otherwise specified (see 6.2), the top coat color shall be camouflage green 383 conforming to MIL-C-46168.

3.18 Fungus and moisture resistance. The electrical circuitry, including all components and connections, except as specified below, shall be protected from the effects of fungus growth and moisture by an overall treatment with a varnish conforming to MIL-V-173, composition I or II, as applicable (see 3.18.1), with one percent copper 8-quinolinolate (by weight) based on the non-volatile content of the varnish.

a. Components or circuit elements that are inherently fungus and moisture resistant or which are hermetically sealed need not be treated,
b. Components or circuit elements whose functions will be adversely affected by the varnish coating shall not be treated.

The varnish shall be applied by spray, brush, or a combination of both to give a dry-film thickness of not less than 1 mil to component or element surfaces previously cleaned and prepared so that the surfaces are free from all foreign matter which would interfere with the adherence or function of the varnish.

3.18.1 Composition. Composition II shall be used only in the case when local air pollution regulations governing the application of varnish precludes the use of composition I. When composition II is used, the contractor shall provide evidence to the Government that the use of composition 11 is required, and shall certify that the composition 11 material complies with Rule 422, South Coast Air Quality Management District.

3.19 Workmanship.

3.19.1 Fabrication. Metal used in the fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the materials to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to ensure uniformity of size and shape.

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3.19.2 Bolted connections. Bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lockwashers shall be provided and all bolts, nuts, and screws shall be tight.

3.19.3 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, or other foreign matter. Weldments, including weld beads, shall be of homogeneous appearance and form without spattering or other irregularities, complying with the size, configuration and other dimensional requirements to develop the full strength of the parts joined by the welds. No cracks of any type are permitted. All welding is to be produced by either AWS or ASME code qualified welders or welding operators and weld shall be in accordance with the same code. Welds shall transmit stress without permanent deformation or failure when the parts connected by the weld are subjected to proof and service loadings. Welds shall be in accordance with one of the codes.

3.19.4 Welders. Before assigning any welder or welding operator to welding work, the contractor shall obtain certification that the welder or welding operator has passed qualification tests as prescribed for the type of welding to be performed by that particular welder or welding operator and that such qualification is effective as defined by the particular code of:

AWS D1.1 - Structural Welding Code - Steel.
AWS D1.2 - Structural Welding Code - Aluminum.
ASME Boiler Pressure Vessel Code, Section IX, Welding Qualifications.

3.19.5 Machine work. Tolerance and gauges for metal fit shall conform to commercial industry standards.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Components and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3 and 4.8,1),
b. Qualilty conformance inspection (see 4.4)
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article inspection. First article inspection shall be performed on one complete unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and test of 4.6.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and the tests of 4.6.

4.5 Examination. Each pumping unit shall be examined as specified herein for the following defects. Presence of one or more defects shall be cause for rejection.

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatments.
105. Used components, pieces, and parts incorporated in the pumping unit.
106. Pump casing not as specified.
107. Pump impeller not as specified.
108. Pump seals not as specified.
109. Screw threads not as specified.
110. Electric motor not as specified.
111. Electric motor assembly and switches not explosion-proof as specified.
112. Slave cable not as specified.
113. Protective frame not as specified.
114. Lifting and tiedowns not as specified.
115. Pump not transportable as specified.
116. Grounding assembly not as specified.
117. Hoses and fittings not as specified.
118. Suction strainer not as specified.
119. Nozzle not as specified.
120. Identification plate not as specified.
121. Treatment or painting not as specified.
122. Electrical circuitry not protected against fungus and moisture as specified.

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123. Workmanship not as specified.
124. Fabrication not as specified.
125. Bolted connections not as specified.
126. Welding not as specified.
127. Welders certification not available.
128. Machine work not as specified.
129. Assembly incorrect or not complete.
130. Parts or components missing or not functioning.

4.6 Tests.

4.6.1 Hydrostatic. The pump and fittings shall be subjected to a hydrostatic test pressure of a minimum of 30 psig for a period of 15 minutes. Any leakage, cracking, or permanent deformation of the assembly shall constitute failure of the test.

4.6.2 Drop test. The pumping unit shall be dropped one time from a height of not less than 48 inches onto a concrete floor. Nonconformance to 3.11 shall constitute failure of this test.

4.6.3 Performance.

4.6.3.1 Test conditions. Unless otherwise specified in a test, test shall be performed in accordance with the applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of a barometric pressure of 29.92 inches of mercury and liquid fuel at 60 °F having a specific gravity of 0.85. Water at a temperature of between 60 °F and 80 °F may be used as the test fluid.

4.6.3.2 Pumping. The pumping unit shall be operated as specified herein to determine pump brake horsepower required, pump efficiency, and the net positive suction head required, based on the data obtained. The resultant data shall be used to plot the corrected performance characteristics.

4.6.3.3 Priming test. The unit shall be tested on the equivalent to a 10-foot static suction lift and shall fully prime within 2 minutes or less. Failure to prime in 2 minutes shall constitute failure of test.

4.6.3.4 Operational test and functional test. The pumping unit shall be operated for 1 hour continuously (as specified in 3.9) at the rated conditions specified in 3.1. The pumping unit shall be operated as required during this period to verify the functional operation of the controls, and the pump shall be examined during the operation for leakage through the pump casing or the shaft seals and for malfunction of any component. Any leakage attributable to defects in design, workmanship, materials, or to the malfunction of any component, or inability of the pump to deliver the minimum capacities specified herein shall constitute failure of this test.

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4.6.3.5 Tilted position. The pumping unit shall be operated for not less than 30 minutes while it is positioned 15 degrees from level in four different directions. Evidence of malfunction or misalignment of components shall constitute failure of this test.

4.6.3.6 Environmental.

4.6.3.6.1 High temperature. The pump shall be tested as specified in MIL-STD-810, method 501.1, procedure I. The maximum ambient storage temperature shall be +155 °F, and the maximum ambient operating temperature shall be +125 °F. The operating period of procedure I shall be 2 hours. The pump shall be operated at rated capacity. Nonconformance to 3.4 shall constitute failure of this test.

4.6.3.6.2 Low temperature. The pump shall be tested as specified in MIL-STD-810, method 502.1, procedure I. The minimum ambient storage temperature shall be -65 °F. The minimum ambient operating temperature shall be -25 °F. The operating period of procedure I shall be 2 hours. The pump shall be operated-at rated capacity. Nonconformance to 3.4 shall constitute failure of this test.

4.6.4 Safety. The pumping unit shall be evaluated for safety requirements throughout testing as specified in 4.6.3. Nonconformance to 3.5 shall constitute failure.

4.6.5 Human factors engineering inspection. The human factors engineering requirements shall be inspected during testing. Nonconformance to 3.6 shall constitute failure.

4.6.6 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474 requirements and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: noise levels shall be measured when equipment is operating under full load, MIL-STD-1474, 5.1.2.1.4. Noise contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with 3.7 provisions shall constitute failure of this test.

4.6.7 Electromagnetic interference. The first article shall be tested to determine conformance with 3.16. Nonconformance to 3.16 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select pumping units at any time during the contract production period and subject these pumping units to the examination specified in 4.5. The inspection will be performed by the Government, at a site selected by the Government, on pumping units selected at random from those which have been accepted by the Government and will not include any previously inspected pumping units. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

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4.7.1 Inspection failure. Failure of an inspection comparison pumping unit to meet any requirement specified herein, during and as a result of the examination and tests specified in 4.6, shall be cause for rejection of the inspection comparison pumping unit and shall be cause for refusal by the Government to continue acceptance of production pumping units until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pumping units previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pumping units accepted prior to the completion of the inspection comparison are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 First article inspection of packaging. The first article packaging shall be examined and tested in accordance with the first article pack (reproduction pack) inspection provisions of MIL-P-10603, and for the additional examination defects specified in 4.8.2.1.

4.8.2 Quality conformance inspection. Packaging of pump assemblies shall be examined in accordance with the quality conformance inspection provisions of MIL-P-10603, and for the additional examination defects specified in 4.8.2.1.

4.8.2.1 Examination. Sample packs, selected in accordance with the sampling requirements of MIL-P-10603, shall be examined for the following defects, in addition to the defects listed in MIL-P-10603. The AQL shall be as specified therein.

Major

131. Commercial packaging not as specified.

5. PACKAGING

5.1 First article packaging. Unless otherwise specified (see 6.2), the contractor shall furnish a first article pack for examination and testing within the time frame specified (see 6.2), to prove prior to the start of production packaging, that the applied preservation, packing, and marking comply with the requirements of this specification. Examination and testing shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may utilize either the first article model or a production model. When the first article model is utilized, and the Government requests a comparison between the first article model and a production model, any preservation and packing shall be removed by the contractor at no expense to the Government.

5.2 Packaging. Packaging of the pumping units shall be level A, level B or commercial as specified (see 6.2).

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5.2.1 Level A and B. Levels A and B packaging, which includes preservation, packing and marking, shall be in accordance with the applicable provisions for levels A and B of MIL-P-10603.

5.2.2 Commercial. Commercial packaging, which includes preservation, packing and marking, shall be in accordance with ASTM D 3951. Additionally, each shipping container, as applicable, shall be marked with the gross weight and cube.

5.3 Depreservation guide. Two depreservation guides shall be prepared. One guide shall be placed in a waterproof envelope marked "Depreservation Guide" and shall be secured to the pump unit in a conspicuous location. The second guide shall be preserved with the technical publications and the bag shall be marked to indicate that the depreservation guide is inside, Unless otherwise specified (see 6.2), DA Form 2258, "Depreservation Guide for Vehicles and Equipment", shall be used.

6. NOTES

6.1 Intended use. The pumping units will be used to transfer fuel from disabled vehicles, under and above ground storage tanks and tank trucks. The unit will operate under severe conditions such as explosion shock, dust and severe climatic elements.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. When a first article is required for the first article inspection and approval (see 3.2, 4.3, and 6.3).
c. When electromagnetic interference suppression is not required (see 3.16).
d. Color of finish coat if not as specified (see 3.17).
e. Time frame required for submission of the first article pack (see 5.1).
f. When first article packaging is not required (see 5.1).
g. Degree of packaging required (see 5.2).
h. When other than DA Form 2258 shall be used (see 5.3).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article.

6.4 First article pack. Any changes or deviations of production packs from the approved first article pack will be subjected to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the pumping unit in accordance with this specification.

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6.5 Depreservation guide. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 5.3)

6.6 Subject term (key word) listing.

35 gallons per minute
Captured fuel
Electric motor driven
Frame mounted
Fuel
Pump

Custodian:
Army - ME

Preparing activity:
Army - ME

Project 4310-A259

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